

MoBiQuant: Problem Setting and Research Gap

Goal: adapt one quantized LLM to changing latency and memory budgets.

Past research: static PTQ

GPTQ, OmniQuant, SmoothQuant, AWQ tune one target bit-width.

They optimize fixed-precision quality, not runtime elasticity.

Changing bit-width usually requires recalibration or quality loss.

Past research: any-precision

AnyPrecisionLLM, AnyBCQ, MatQuant support multiple precisions.

But they add scaling, repacking, table lookups, or kernel overhead.

Switching precision is not yet cheap at token granularity.

What is missing

A single nested model representation.

Token-level precision decisions during inference.

A low-overhead kernel path that fetches only needed bits.

MoBiQuant frames precision as a token-level runtime resource



Core Observation: Outlier Migration

The tokens that dominate quantization error move when the bit-width changes.

Why static calibration fails

Calibration parameters overfit the outlier distribution of one bit-width.

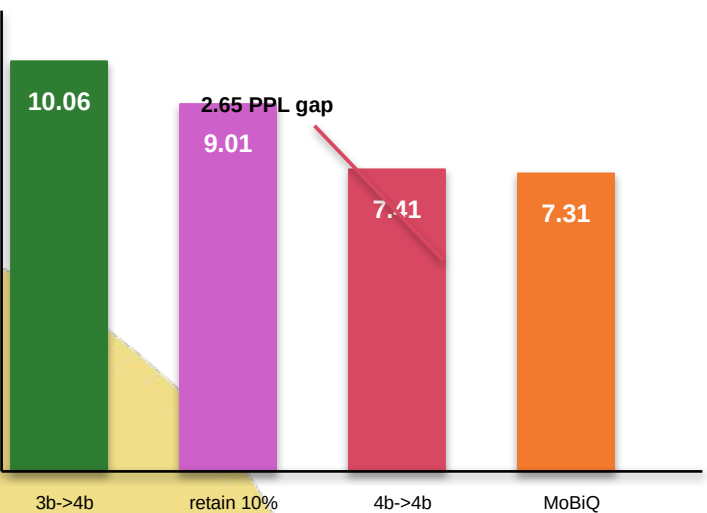
A token well-fitted at 4-bit can become an outlier at 3-bit.

What MoBiQuant exploits

Token sensitivity can guide per-token bit allocation.

Routing high-sensitivity tokens through more bit slices improves generalization.

LLaMA3-8B / WikiText2 PPL

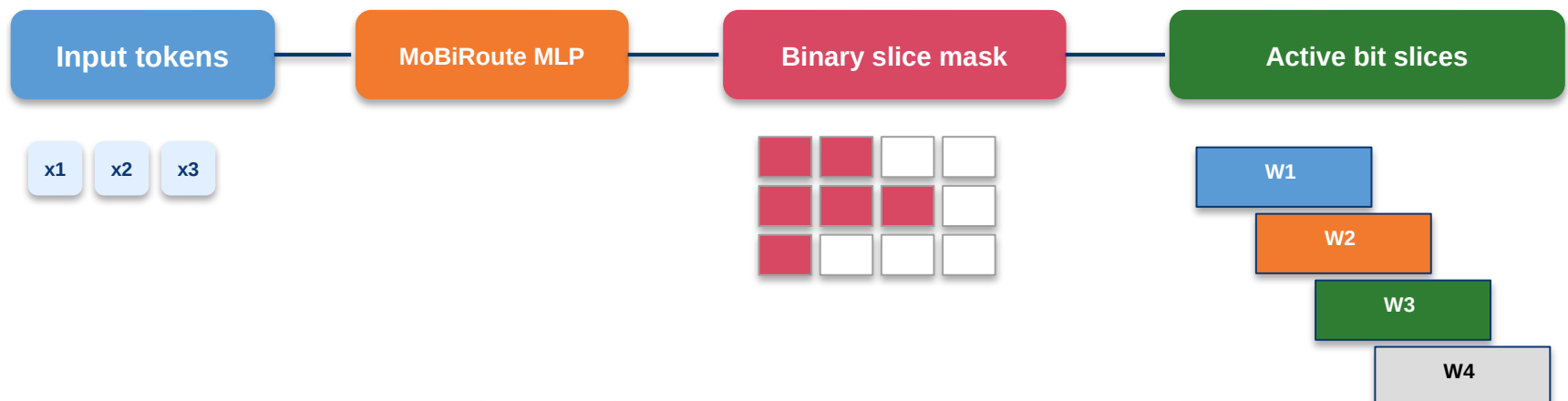


Token-wise error peaks shift across precision



Method: MoBiSlice and MoBiRoute

A single linear block becomes a routed mixture of residual bit slices.



MoBiSlice

Recursively quantizes residual weights into 2-bit slices.
Higher precision is reconstructed by summing more slices.

MoBiRoute

Scores token sensitivity and gates slices with binary decisions.
A threshold controls the target average bit budget.

Kernel design

Bit-major packing fetches only active slices.
Shared scaling avoids extra per-precision parameters.



Main Results and Takeaways

MoBiQuant keeps elasticity while approaching fixed-precision PTQ quality.

Accuracy

Matches or surpasses static scalar PTQ across LLaMA2 and LLaMA3 model families.
At 2-3 bit, reduces severe PPL collapse versus prior any-precision baselines.

Throughput

Average decoding speedup: 33.8% over AnyPrecisionLLM.
Average decoding speedup: 22.8% over AnyBCQ.

Memory

One nested model replaces multiple per-bit deployments.
Reported memory footprint reduction is up to 3.5x.

Conclusion

The core win is not just storing more precisions.
The router mitigates outlier migration by assigning precision per token.

Takeaway: precision becomes a dynamic information budget at inference time

